

A METHOD FOR SELECTING THE COVARIATES IN ANALYSIS OF COVARIANCE

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Summary: Two types of error are involved when a statistical model is fitted to data; firstly the family of models which is fitted contains, in general, only an approximation to the true situation and secondly there are errors due to sampling, i.e. the parameters can only be estimated.

A method of "discrepancies" which attempts to take both of these types of error into account has been applied by the authors to a number of statistical problems. In this paper we show how the method can be applied to decide on which covariates, if any, should be retained in an analysis of covariance model.

1. MODEL SELECTION BY EXPECTED DISCREPANCIES

Certain basic ideas on model selection were applied by the authors to various statistical problems such as the fitting of distributions, analysis of contingency tables, choice of linear models, spectral analysis, fitting of models to station-

ary time series, etc. These ideas are described heuristically in chapter 12 of Linhart and Zucchini (1980), in more detail in (1982a) and are applied to elementary problems in (1982b). In this paper we apply them to the problem of model selection in analysis of covariance.

The ideas were developed over many years of experience in analysing data. In certain areas of Statistics they have been implicitly used for many decades and in the recent model building literature some of them appear explicitly. It would be difficult to give a list of references which does justice to everybody. We thank the referee for bringing Sawa (1978) to our notice in which the ideas of Akaike (1973) are clearly expressed. In this context White (1982) is also relevant. Akaike's information criterion is, however, only one example of methods which emerged on the basis of such or similar ideas. Each of the more sophisticated variable selection criteria belongs to this category. The work of Parzen (see, e.g., 1974), Wahba (see, e.g., 1977), Bunke and collaborators (see, e.g., Bunke and Schmidt (1980)) must also be mentioned.

We believe that in most applications of statistics the mechanisms which generate the data are too complex to be represented exactly by a model, i.e. there can be no "true model". At best one can find a model which generates numbers which have "almost" the same structure as the data. We call such models "operating models" and assume that in a given situation it is possible to find a reasonable operating model, ν , which is in principle estimable.

On practice however even an operating model can either be too complex to be used at all or it may contain more parameters than can be reasonably estimated. Consequently a simpler model from a family $\mu_\theta, \theta \in \Theta$ has to be fitted. The quality of the fit can be judged by a discrepancy $\Delta(\mu_\theta, \nu) \geq 0$, a functional on the square of all probability distributions, the choice of which depends on the purpose of the analysis.

The functional Δ is to be chosen in such a way that a decrease in its value implies an improved fit.

If a unique θ_0 exists such that $\theta_0 = \arg \min_{\theta \in \Theta} \Delta(\mu_\theta, \nu)$

we call $\Delta(\mu_{\theta_0}, \nu)$ the discrepancy due to approximation.

If $\hat{\theta}$ is an estimator of θ_0 we call $\Delta(\mu_{\hat{\theta}}, \mu_{\theta_0})$ the discrepancy due to estimation and $\Delta(\mu_{\hat{\theta}}, \nu)$ the overall discrepancy.

The suitability of the family $\mu_\theta, \theta \in \Theta$ and of the estimation method is measured by the expected overall discrepancy $E_\nu \Delta(\mu_{\hat{\theta}}, \nu)$, which we will refer to simply as the expected discrepancy. An estimator of the expected discrepancy is called a criterion.

The method which we propose is to select the family of models which leads to the smallest value of the criterion.

In analysis of variance with the operating model given by

$$y = A\theta + e$$

where A is a known $(n \times p)$ matrix of rank $p < n$, θ a vector of parameters and e a random vector with $Ee = 0$ and $\text{Var } e = \Sigma$, Σ nonsingular and unknown, a reasonable discrepancy is the sum of squared differences between the expectations of the observations under the fitted and operating model; i.e.

$$\Delta(\mu_\theta, \nu) = (E_{\mu_\theta} y - E_\nu y)' (E_{\mu_\theta} y - E_\nu y).$$

The fitted model here is the operating model with some of the effects left out. If one uses the customary estimators for the parameters and the analysis is carried out according to the method described above then the following result emerges (Linhart and Zucchini, 1982a):

Eliminate a certain effect from the operating model if the corresponding F-statistic (used to test the hypothesis that the effect is zero in the usual analysis of variance table) is less than 2.

It is also possible to test the hypothesis that the expected discrepancy increases if a certain effect is not eliminated from the operating model. To be able to do this however it is necessary to assume homoscedasticity, normality and independence of all random variables in the equation of the operating model. Here one uses the same F-statistic, but a new table of critical values based on the noncentral F-distribution.

The discrepancy which was used in this case is the sum of squared differences between operating and fitted expectations. Its use is justified since in analysis of variance the operating expectation is in the centre of interest. If one interprets Mallows' (1973) C_p , or rather a statistic which is equivalent to it (cf., e.g., Thompson (1978)), as estimator of an expected discrepancy, the discrepancy which is at the basis of the method is the average of the mean squared errors of prediction, where the average is taken over the predictor sets in the regression sample. One can prove that this discrepancy is *equivalent* to the discrepancy which we used for linear models and analysis of variance in particular. It follows immediately that the decision rule which emerges for *fixed* analysis of variance models with independent and homoscedastic errors by applying C_p is the same as the rule obtained here, in spite of the fact that the logic which lead to the choice of a discrepancy is *different* to that employed here. What really emerges is that the C_p -discrepancy (which has an eye on good *predictions*) is actually a discrepancy of *internal* goodness of fit of the expectation. (The reason for this is that the inconvenient dependence of the mean squared error of prediction on the predictor set is overcome by averaging

over the predictor sets *in the regression sample.*)

Toro-Wizcarrondo and Wallace (1968) gave a test in regression analysis which is in the case of *fixed* analysis of variance models the test described here. Browne (1969) introduced a test in connection with the C_p -criterion which is (because of the mentioned equivalence of the employed discrepancies) also identical to our test for fixed analysis of variance models.

2. A DISCREPANCY FOR ANALYSIS OF COVARIANCE

Although the particular discrepancy which was used for analysis of variance can also be used for linear models in general it is not always appropriate for analysis of covariance.

In some applications of analysis of covariance the covariates are of interest in themselves. However in many applications covariates are introduced primarily to increase the precision of the estimates for the treatment effects. In the first case the discrepancy mentioned above would be suitable, but in the second case the discrepancy should be sensitive to treatment effects only, and not to the influence of the covariates on the expectation of the observations. Here a particular covariate should be included in the model only if it leads to better estimates of the treatment effects.

Let the operating model be

$$y = A_1 \theta_1^* + A_2 \theta_2^* + A_3 \theta_3^* + e$$

where y is a column vector of n observations, A_1 , A_2 and A_3 are known matrices of dimension $n \times I$, $n \times b$ and $n \times (p-b)$, respectively, ($b = 0, 1, 2, \dots, p$), e is a random vector with $Ee = 0$ and $\text{Var } e = \sigma^2 I$, θ_1^* is a vector of I treatment parameters and θ_2^* and θ_3^* are vectors of regression coefficients of the covariates. (We use the stars here to indicate

that the model is fully specified. Later if no confusion can arise the stars will be omitted.) We assume that there are equal numbers of observations for each treatment.

Suppose that we wish to decide whether the estimate of θ_1^* is improved if the p-b covariates with coefficients θ_3^* in the operating model are omitted, i.e. if the family of models to be fitted is

$$y = A_1 \theta_1 + A_2 \theta_2 + e.$$

A possible discrepancy is then

$$\Delta(\mu_\theta, \nu) = (\theta_1^* - \theta_1)'(\theta_1^* - \theta_1).$$

The discrepancy due to approximation is zero, but each different value of b leads to a different estimator $\hat{\theta}_1$ and hence to a different discrepancy due to estimation and a different overall discrepancy. The expected overall discrepancy is given by

$$E(\hat{\theta}_1 - \theta_1^*)'(\hat{\theta}_1 - \theta_1^*)$$

where $\hat{\theta}_1$ is an estimator of θ_1^* using the family of models being fitted.

3. THE EXPECTED DISCREPANCY

The following notation is used:

$$A = (A_1 \ A_2 \ A_3), \quad A_{12} = (A_1 \ A_2)$$

$$C_{ij} = A_i' A_j, \quad C = \{C_{ij}\}, \quad C^{-1} = \{C^{ij}\}$$

$$V = \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix}, \quad V^{-1} = \begin{bmatrix} v^{11} & v^{12} \\ v^{21} & v^{22} \end{bmatrix}$$

$$C_3 = (C_{31} \ C_{32}) , \quad W = (V^{11} \ V^{12}) ,$$

$$\theta' = (\theta_1' \ \theta_2' \ \theta_3') .$$

The least squares estimator of θ_1 using the (reduced) fitted model is

$$\hat{\theta}_1 = W A'_{12} Y .$$

The expectation and variance of $\hat{\theta}_1$ under the operating model are given by

$$E\hat{\theta}_1 = W A'_{12} A \theta \text{ and } \text{Var } \hat{\theta}_1 = \sigma^2 V^{11} .$$

It is easy to show that

$$(E\hat{\theta}_1 - \theta_1)'(E\hat{\theta}_1 - \theta_1) = \theta_3' C_3 W' W C_3' \theta_3$$

and so the expected discrepancy is given by

$$\begin{aligned} E(\hat{\theta}_1 - \theta_1)'(\hat{\theta}_1 - \theta_1) &= (E\hat{\theta}_1 - \theta_1)'(E\hat{\theta}_1 - \theta_1) + \text{tr Var } \hat{\theta}_1 \\ &= \theta_3' C_3 W' W C_3' \theta_3 + \sigma^2 \text{tr } V^{11} . \end{aligned}$$

This result holds for all $b = 0, 1, 2, \dots, p$. If the family containing the operating model is used for fitting (i.e. $b = p$, $V = C$) then $E\hat{\theta}_1 = \theta_1$ and the expected discrepancy reduces to $\sigma^2 \text{tr } C^{11}$.

A family with $b < p$ covariates should be fitted in preference to the operating model if the difference in expected discrepancies is negative, i.e. if

$$\sigma^2 \text{tr } (V^{11} - C^{11}) + \theta_3' C_3 W' W C_3' \theta_3 < 0 .$$

4. THE CRITERION

Since $\hat{\theta}_3' C_3 W' W C_3' \hat{\theta}_3$, with $\hat{\theta}_3 = (C^{31} C^{32} C^{33}) A' y$ has expectation

$$\theta_3' C_3 W' W C_3' \theta_3 + \sigma^2 \text{tr} C_3 W' W C_3' C^{33}$$

it follows that an unbiased estimator for the difference in expected discrepancies is given by

$$\hat{\sigma}^2 \text{tr} (V^{11} - C^{11}) + \hat{\theta}_3' C_3 W' W C_3' \hat{\theta}_3 - \hat{\sigma}^2 \text{tr} C_3 W' W C_3' C^{33}$$

where $\hat{\sigma}^2$ is the residual mean square when the operating model is fitted, i.e.

$$\hat{\sigma}^2 = (y'y - \hat{\theta}' C \hat{\theta}) / (n - I - p).$$

One can prove that $\text{tr} C_3 W' W C_3' C^{33} = \text{tr} (C^{11} - V^{11})$, and so the criterion for the reduced family of models is smaller if

$$\hat{\theta}_3' C_3 W' W C_3' \hat{\theta}_3 - 2 \hat{\sigma}^2 \text{tr} (C^{11} - V^{11}) < 0.$$

Clearly, the above results can also be applied to decide whether b_1 or b_2 , where $b_1 < b_2 < p$, covariates should be used. The family leading to the smaller value of the criterion should be fitted.

5. SPECIAL CASES

The case of one covariate only ($b = 0$, $p = 1$)

In this case the results can be expressed in terms of the usual quantities which are calculated in analysis of covariance. We use the notation

$$y_{ij} = \mu_i + \theta_3 x_{ij} + e_{ij}, \quad i=1,2,\dots,I, \quad j=1,2,\dots,J.$$

Source of variation	d.f.	SS_{yy}	SP_{yx}	SS_{xx}
Groups (G)	I	$\sum_{ij} \bar{y}_{i.}^2$	$\sum_{ij} \bar{y}_{i.} \bar{x}_{i.}$	$\sum_{ij} \bar{x}_{i.}^2$
Error (E)	$n - I$	$\sum_{ij} (y_{ij} - \bar{y}_{i.})^2$	$\sum_{ij} (y_{ij} - \bar{y}_{i.})(x_{ij} - \bar{x}_{i.})$	$\sum_{ij} (x_{ij} - \bar{x}_{i.})^2$
Total (T)	$n = IJ$	$\sum_{ij} y_{ij}^2$	$\sum_{ij} y_{ij} x_{ij}$	$\sum_{ij} x_{ij}^2$

Here $\hat{\theta}_3 = SP_{yx}(E) / SS_{xx}(E)$,

$$\hat{\theta}_3' C_3 W' W C_3' \hat{\theta}_3 = \hat{\theta}_3^2 SS_{xx}(G) / J,$$

$$\text{tr}(C^{11} - V^{11}) = (SS_{xx}(G) / SS_{xx}(E)) / J.$$

The criterion is

$$(SS_{xx}(G) / SS_{xx}(E)) (SP_{yx}^2(E) / SS_{xx}(E) - 2\hat{\theta}_3^2)$$

and so the covariate should be omitted if

$$F = (SP_{yx}^2(E) / SS_{xx}(E)) / \hat{\theta}_3^2 < 2$$

$$\text{where } \hat{\theta}_3^2 = (SS_{yy}(E) - SP_{yx}^2(E) / SS_{xx}(E)) / (IJ - I - 1).$$

The above result does not depend on the assumption that the errors are normally distributed. If one assumes that this does hold then it is also possible to test the hypothesis that the difference in expected discrepancies is negative, i.e. the hypothesis that

$$E \hat{\theta}_3' C_3 W' W C_3' \hat{\theta}_3 - 2\hat{\theta}_3^2 \text{tr}(C^{11} - V^{11}) < 0, \quad \text{i.e. that}$$

$$E(SP_{yx}^2(E) / SS_{xx}(E)) / \sigma^2 < 2.$$

It is well known that under normality the sum of squares for the regression coefficient of the covariate, $SS_{yx}^2(E) / S_{xx}(E)$, is distributed as $\sigma^2 \chi^2(1, \lambda)$.

The hypothesis to be tested is thus

$$E\sigma^2 \chi^2(1, \lambda) / \sigma^2 < 2, \text{ i.e. } \lambda < 1.$$

As test statistic one can use F which, under the operating model, has the noncentral F -distribution $F(\nu_1 = 1, \nu_2 = n-1; \lambda)$. The upper percentage points of the noncentral F with $\lambda = \nu_1 = 1$ are tabulated in Linhart and Zucchini (1982b). The hypothesis has to be rejected and the covariate has to be used if the observed F is larger than the tabulated critical value.

In this case (only) the method which emerges is equivalent to that obtained for analysis of variance. This is in spite of the fact that a different discrepancy is used.

The rule of using a critical value of 2 for F corresponds again roughly to a test of the mentioned hypothesis at a significance level of approximately 0.5.

The case $b = 0, p > 1$

For this case too the results can be somewhat simplified. Let $SS_{x_1x_1}, SP_{x_1x_2}, SP_{yx_1}$, etc. be defined analogously to the sums of squares and products in the previous case and let

$$S = \begin{bmatrix} SS_{x_1x_1} & \dots & SP_{x_1x_p} \\ \vdots & & \\ SS_{x_px_1} & \dots & SS_{x_px_p} \end{bmatrix}$$

$$S'_{yx} = (SP_{yx_1}, \dots, SP_{yx_p}).$$

Here one obtains

$$\begin{aligned} \text{tr}(C^{11} - V^{11}) &= \text{tr} S(G) S^{-1}(E) / J, \quad C_3 W' W C_3' = S(G) / J \quad \text{and} \\ \hat{\theta}_3 &= S^{-1}(E) S_{yx}(E). \end{aligned}$$

The estimated difference of expected discrepancies is

$$(\hat{\theta}_3' S(G) \hat{\theta}_3 - 2 \hat{\sigma}^2 \text{tr} S(G) S^{-1}(E)) / J$$

which is negative if

$$H = (\hat{\theta}_3' S(G) \hat{\theta}_3 / \text{tr} S(G) S^{-1}(E)) / \hat{\sigma}^2 < 2.$$

An exact test of the hypothesis that the difference in expected discrepancies is negative cannot be carried out by means of H . One can prove that the numerator of H is independently distributed of σ^2 (the mean square error) but it does not have a Chi-squared distribution. The mean and variance of the quadratic form were calculated using the characteristic function of quadratic forms in normal variables which is given, e.g., in Plackett (1960):

$$E \hat{\theta}_3' S(G) \hat{\theta}_3 = \theta_3' S(G) \theta_3 + \sigma^2 \text{tr} S(G) S^{-1}(E)$$

$$\begin{aligned} \text{Var } \hat{\theta}_3' S(G) \hat{\theta}_3 &= 4 \sigma^2 \theta_3' S(G) S^{-1}(E) S(G) \theta_3 \\ &\quad + 2 \sigma^4 \text{tr} (S(G) S^{-1}(E))^2. \end{aligned}$$

6. AN EXAMPLE OF APPLICATION

The following data, taken from Snedecor and Cochran (1980), show the initial weight in lb, x_1 , the initial age in days, x_2 , and the average weight gain in lb/day, y , of 40 pigs subjected to 4 different treatments.

Treatment 1		
x_1	x_2	y
61	78	1,40
59	90	1,79
76	94	1,72
50	71	1,47
61	99	1,26
54	80	1,28
57	83	1,34
45	75	1,55
41	62	1,57
40	67	1,26

Treatment 3		
x_1	x_2	y
80	78	1,67
61	83	1,41
62	79	1,73
47	70	1,23
59	85	1,49
42	83	1,22
47	71	1,39
42	66	1,39
40	67	1,56
40	67	1,36

Treatment 2		
x_1	x_2	y
74	78	1,61
75	99	1,31
64	80	1,12
48	75	1,35
62	94	1,29
42	91	1,24
52	75	1,29
43	63	1,43
50	62	1,29
40	67	1,26

Treatment 4		
x_1	x_2	y
62	77	1,40
55	71	1,47
62	78	1,37
43	70	1,15
57	95	1,22
51	96	1,48
41	71	1,31
40	63	1,27
45	62	1,22
39	67	1,36

The sums of squares and products are given by

SOURCE OF VARIATION	d.f.	$SS_{x_1x_1}$	$SP_{x_1x_2}$	$SS_{x_2x_2}$
Groups (G)	4	111386,1	162658,6	237655,8
Error (E)	36	4876,9	2877,4	4548,2
TOTAL (T)	40	116263,0	165536,0	242204,0

SOURCE OF VARIATION	d.f.	SS_{yy}	SP_{yx_1}	SP_{yx_2}
Groups (G)	4	77,2671	2929,141	4279,887
Error (E)	36	0,8452	26,219	5,623
TOTAL (T)	40	78,1123	2955,360	4285,510

A conventional analysis of covariance reveals that the covariate x_1 is significant at the 5% level but that x_2 is not. A rule of thumb, apparently due to W.G. Cochran, is to include a covariate in the model if the "correlation coefficient"

$$\rho = SP_{yx}(E) / \sqrt{SS_{xx}(E) SS_{yy}(E)}$$

is greater than 0,3. In this example the values of ρ for x_1 and x_2 are approximately 0,4 and 0,1 respectively. Although this rule was probably intended for cases where there is a single covariate these results suggest that only x_1 , the initial weight, should be retained in the model.

It is easy to show that for the case of a single covariate the rule described in section 5 reduces to: Include the covariate in the model if $\rho > \sqrt{2 / (IJ - I + 1)}$. (For $I = 4$ and $J = 10$ this "critical value" is 0,23.)

We now apply the results derived above to this data, firstly to

decide whether it is preferable to omit both covariates from the model, i.e. whether $b=0$ leads to a smaller criterion than the operating model.

Here

$$S(E) = \begin{bmatrix} 4876,9 & 2877,4 \\ 2877,4 & 4548,2 \end{bmatrix}, \quad S^{-1}(E) = 10^{-4} \begin{bmatrix} 3,2717 & -2,0698 \\ -2,0698 & 3,5081 \end{bmatrix}$$

$$S(G) = \begin{bmatrix} 111386,1 & 162658,6 \\ 162658,6 & 237655,8 \end{bmatrix}, \quad S_{yx}(E) = \begin{bmatrix} 26,219 & 5,623 \end{bmatrix}$$

$$\text{Hence } \hat{\theta}_3 = (0,007414 \quad -0,003454) \text{ and } \hat{\sigma}^2 = 0,019713.$$

The estimated difference in expected discrepancies is then $(0,62716 - 2 \times 0,01971 \times 52,48) / 10 = -0,144$. Since this is *negative* one should *not* use both variables.

It is conceivable, however, that the use of the single covariate x_1 is preferable to using no covariates at all. To decide on this we compare at first the model containing both covariates to that having only x_1 .

One needs to compute C^{-1} and V^{-1} . With these inverses one can then compute

$$\begin{aligned} \text{tr}(C^{11} - V^{11}) &= 2,964, \quad C_2 W' W C_3' = 7768,7, \\ \hat{\theta}_3 &= -0,003457 \text{ and } \hat{\sigma}^2 = 0,019713. \end{aligned}$$

The difference between the two criteria is therefore

$$(-0,003457)^2 \times 7768,7 - 2 \times 2,964 \times 0,01971 = -0,0240.$$

The difference is *negative* and so the model with the single covariate x_1 is to be preferred to that with both covariates.

Since $-0,144 < -0,024$ one would decide to use no covariate at all.

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